

## SEMESTER S5

### MACHINE LEARNING

(Common to CS/AD/CR/CA/CC/CD)

|                                            |                 |                    |                |
|--------------------------------------------|-----------------|--------------------|----------------|
| <b>Course Code</b>                         | <b>PCCST503</b> | <b>CIE Marks</b>   | 40             |
| <b>Teaching Hours/Week<br/>(L: T:P: R)</b> | 3:0:0:0         | <b>ESE Marks</b>   | 60             |
| <b>Credits</b>                             | 3               | <b>Exam Hours</b>  | 2 Hrs. 30 Min. |
| <b>Prerequisites (if any)</b>              | None            | <b>Course Type</b> | Theory         |

#### Course Objectives:

1. To impart the fundamentals principles of machine learning in computer and science.
2. To provide an understanding of the concepts and algorithms of supervised and unsupervised learning.

### SYLLABUS

| <b>Module No.</b> | <b>Syllabus Description</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Contact Hours</b> |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>1</b>          | <b>Introduction to ML :-</b><br>Machine Learning vs. Traditional Programming, Machine learning paradigms - supervised, semi-supervised, unsupervised, reinforcement learning.<br>Parameter Estimation - Maximum likelihood estimation (MLE) and maximum a posteriori estimation (MAP), Bayesian formulation.<br><b>Supervised Learning :-</b><br>Feature Representation and Problem Formulation, Role of loss functions and optimization<br>Regression - Linear regression with one variable, Linear regression with multiple variables : solution using gradient descent algorithm and matrix method. | <b>9</b>             |
| <b>2</b>          | <b>Classification</b> - Logistic regression, Naïve Bayes, KNN, Decision Trees – ID3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>9</b>             |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                       |          |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|          | <p>Generalisation and Overfitting - Idea of overfitting, LASSO and RIDGE regularization, Idea of Training, Testing, Validation</p> <p>Evaluation measures – Classification - Precision, Recall, Accuracy, F-Measure, Receiver Operating Characteristic Curve(ROC), Area Under Curve (AUC).</p> <p>Regression - Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), R Squared/Coefficient of Determination.</p> |          |
| <b>3</b> | <p>SVM – Linear SVM, Idea of Hyperplane, Maximum Margin Hyperplane, Non-linear SVM, Kernels for learning non-linear functions</p> <p>Neural Networks (NN) - Perceptron, Neural Network - Multilayer feed-forward network, Activation functions (Sigmoid, ReLU, Tanh), Back propagation algorithm.</p>                                                                                                                 | <b>9</b> |
| <b>4</b> | <p><b>Unsupervised Learning</b></p> <p>Clustering - Similarity measures, Hierarchical Clustering - Agglomerative Clustering, partitional clustering, K-means clustering</p> <p>Dimensionality reduction - Principal Component Analysis, Multidimensional scaling</p> <p>Ensemble methods - bagging, boosting; Resampling methods - Bootstrapping, Cross Validation. Practical aspects - Bias-Variance tradeoff.</p>   | <b>9</b> |

**Course Assessment Method**  
(CIE: 40 marks, ESE: 60 marks)

**Continuous Internal Evaluation Marks (CIE):**

| <b>Attendance</b> | <b>Assignment/<br/>Microproject</b> | <b>Internal<br/>Examination-1<br/>(Written)</b> | <b>Internal<br/>Examination- 2<br/>(Written )</b> | <b>Total</b> |
|-------------------|-------------------------------------|-------------------------------------------------|---------------------------------------------------|--------------|
| <b>5</b>          | <b>15</b>                           | <b>10</b>                                       | <b>10</b>                                         | <b>40</b>    |

**End Semester Examination Marks (ESE)**

*In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions*

| <b>Part A</b>                                                                                                                                                          | <b>Part B</b>                                                                                                                                                                                                                                                                         | <b>Total</b> |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| <ul style="list-style-type: none"> <li>• 2 Questions from each module.</li> <li>• Total of 8 Questions, each carrying 3 marks</li> </ul> <p><b>(8x3 =24 marks)</b></p> | <ul style="list-style-type: none"> <li>• Each question carries 9 marks.</li> <li>• Two questions will be given from each module, out of which 1 question should be answered.</li> <li>• Each question can have a maximum of 3 subdivisions.</li> </ul> <p><b>(4x9 = 36 marks)</b></p> | <b>60</b>    |

### Course Outcomes (COs)

At the end of the course students should be able to:

| <b>Course Outcome</b> |                                                                                 | <b>Bloom's Knowledge Level (KL)</b> |
|-----------------------|---------------------------------------------------------------------------------|-------------------------------------|
| <b>CO1</b>            | Illustrate Machine Learning concepts and basic parameter estimation methods.    | <b>K2</b>                           |
| <b>CO2</b>            | Demonstrate supervised learning concepts (regression, classification).          | <b>K3</b>                           |
| <b>CO3</b>            | Illustrate the concepts of Multilayer neural network and Decision trees         | <b>K3</b>                           |
| <b>CO4</b>            | Describe unsupervised learning concepts and dimensionality reduction techniques | <b>K3</b>                           |
| <b>CO5</b>            | Use appropriate performance measures to evaluate machine learning models        | <b>K3</b>                           |

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

### CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

|            | <b>PO1</b> | <b>PO2</b> | <b>PO3</b> | <b>PO4</b> | <b>PO5</b> | <b>PO6</b> | <b>PO7</b> | <b>PO8</b> | <b>PO9</b> | <b>PO10</b> | <b>PO11</b> | <b>PO12</b> |
|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|-------------|-------------|-------------|
| <b>CO1</b> | 3          | 3          | 3          |            |            |            |            |            |            |             |             | 3           |
| <b>CO2</b> | 3          | 3          | 3          |            |            |            |            |            |            |             |             | 3           |
| <b>CO3</b> | 3          | 3          | 3          |            |            |            |            |            |            |             |             | 3           |
| <b>CO4</b> | 3          | 3          | 3          |            |            |            |            |            |            |             |             | 3           |
| <b>CO5</b> | 3          | 3          | 3          |            |            |            |            |            |            |             |             | 3           |

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

| Text Books |                                                               |                                 |                            |                  |
|------------|---------------------------------------------------------------|---------------------------------|----------------------------|------------------|
| Sl. No     | Title of the Book                                             | Name of the Author/s            | Name of the Publisher      | Edition and Year |
| 1          | Introduction to Machine Learning                              | Ethem Alpaydin                  | MIT Press                  | 4/e, 2020        |
| 2          | Data Mining and Analysis: Fundamental Concepts and Algorithms | Mohammed J. Zaki   Wagner Meira | Cambridge University Press | 1/e, 2016        |
| 3          | Neural Networks for Pattern Recognition                       | Christopher Bishop              | Oxford University Press    | 1/e, 1998        |

| Reference Books |                                       |                                     |                       |                  |
|-----------------|---------------------------------------|-------------------------------------|-----------------------|------------------|
| Sl. No          | Title of the Book                     | Name of the Author/s                | Name of the Publisher | Edition and Year |
| 1               | Applied Machine Learning              | M Gopal                             | McGraw Hill           | 2/e, 2018        |
| 2               | Machine Learning using Python         | Manaranjan Pradhan   U Dinesh Kumar | Wiley                 | 1/e, 2019        |
| 3               | Machine Learning: Theory and Practice | M.N. Murty, V.S. Ananthanarayana    | Universities Press    | 1/e, 2024        |

| Video Links (NPTEL, SWAYAM...) |                                                                                                                             |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| No.                            | Link ID                                                                                                                     |
| 1                              | <a href="https://archive.nptel.ac.in/courses/106/105/106105152/">https://archive.nptel.ac.in/courses/106/105/106105152/</a> |
| 2                              | <a href="https://archive.nptel.ac.in/courses/106/106/106106139/">https://archive.nptel.ac.in/courses/106/106/106106139/</a> |
| 3                              | <a href="https://nptel.ac.in/courses/106106202">https://nptel.ac.in/courses/106106202\</a>                                  |